CSE3413 Software Testing
Sem II 2025/26

Software Test Plan

«DobiDemo»

Prepared by :
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Document Information

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Project No. : 14
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Issue Date : 13.04.2026

Document History

Version	Date	Summary of change	Author
1.0	13 April 2026	Release version	

Approvals

Name	Role	Signature	Escalation
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1
Introduction

This part provides a high-level overview of the testing objectives, scope, context, and the significance of the project. It sets the stage for the rest of the document by providing essential background information and the rationale for the testing effort.

1.1 Purpose

The purpose of this test plan is to outline the testing strategy, approach, resources, and schedule for testing the DobiDemo Laundry Management application to ensure it meets the specified requirements and functions as expected.

1.2 Project Overview

The Dobi Demo Laundry Management System is a website used to manage laundry business operations. The system can be used for user login, managing customer information, handling orders, checking laundry progress, processing payments, and managing branch information.

1.3 Scope

Define the boundaries of the testing effort. Specify what will be tested and what will not be tested.

1.3.1 Features to be Tested

This test plan covers three main modules which are the Product Module, Supplier Module, and Customer Module. The Product Module allows users to add, search, view, and manage product information including product name, unit, price, and stock details. The Supplier Module is used to manage supplier information such as supplier name, contact details, account balance, and supplier records. The Customer Module handles customer data management, including adding new customers, viewing customer details, sorting customer records, categorizing customers based on sales activity, and managing customer account information.

1.3.2 Features Not to be Tested

This test plan does not cover server hosting infrastructure, source code internal review, third-party payment gateway security, and mobile application versions if available. Additionally, system deployment configurations, database server performance tuning, and external API services are outside the scope of this testing plan and will not be tested in this phase.

1.4 Roles and Responsibilities

This section outlines the roles involved in the testing process and their respective responsibilities to ensure clarity and accountability.

1.4.1 Roles

No.	Name	Role
1.	Dr Ahmad Muhaimin Bin Ismail	Project Leader
2.	Nurul Alessa Binti Mohd Nizam	Project Manager (PM)
3.	Nurfatihah Nadziha Binti Bakri	Test Analyst (TA)
4.	Balqis Nur Insyirah Binti Mohd Sharif	Test Engineer (TE)

1.4.2 Responsibilities

No.	Task	PM	TA	TE
1.	Prepare Software Test Plan	X		
2.	Design test cases	X	X	X
3.	Perform manual testing		X	X
4.	Perform automation testing		X	X
5.	Report and track defects		X	X
6.	Prepare documentation	X		

1.5 Definitions, Acronyms, and Abbreviations

Provide definitions for any technical terms or acronyms used in the test plan.

Term/Acronym	Definition
QA	Quality Assurance
UAT	User Acceptance Testing
SUT	System Under Test

1.6 References

*T*his test plan references the following documents :

1. GeeksforGeeks. (2026, April 2). *Introduction to software testing*. GeeksforGeeks. Available at : https ://www.geeksforgeeks.org/software-testing/software-testing-basics/
2. CSE3413 DobiDemo System Description v1.0.

2
Test Strategy

This part outlines the overall approach and methodology that will be used to conduct the testing. It serves as a guide for how the testing will be performed, ensuring consistency and comprehensiveness in the testing process.

2.1 Testing Types

2.1.1 Testing Types

This section outlines the various types of testing that will be performed to ensure the quality, functionality, and reliability of the DobiDemo system.

— **Static Testing :** Static testing is conducted without executing the system. It involves reviewing system requirements, test documentation, and design components to identify defects at an early stage. In this project, technical review is used to ensure that requirements, test cases, and system structure are clear, consistent, and complete.

— **Dynamic Testing :** Dynamic testing is performed by executing the system to evaluate its actual behavior and functionality.

  — **Black-box Testing :** This technique validates system functionality by focusing on inputs and expected outputs without considering internal implementation. It is used to verify features such as data input, transaction processing, and system calculations.

  — **Experience-Based Testing :** This technique relies on tester experience and intuition to discover defects that may not be covered by predefined test cases. It includes trying unusual inputs, random navigation, and observing unexpected system behavior.

2.1.2 Functional Testing

Module	Functional Requirement
Customers	[FR-101] User must be able to add a new customer with name, phone number, description and starting balance (if any). [FR-102] User must be able to view a list of customers, sort them according to Name, category or Account Balance and search for customers by name. [FR-103] User must be able to view customer details including name, phone number, category, description, and current account balance. The system must be able to show all sales related to the customer. [FR-104] System must be able to categorize customer based on number of sales. [FR-105] User must be able to update existing customer details
Supplier	[FR-301] User must be able to add a new supplier with name, phone number, description and starting balance (if any). [FR-302] User must be able to view a list of suppliers, sort them according to Name or phone number and search for supplier by name. [FR-303] User must be able to view supplier details including name, phone number, description, and current account balance. The system must be able to list all the expenses related to the supplier. [FR-304] User must be able to update existing supplier details.
Products	[FR-501] User must be able to add new products with name, measurement unit, and unit price. [FR-502] User must be able to view a list of products, sort them according to Name or price per unit and search for the product by name. [FR-503] User must be able to view product details. [FR-504] User must be able to update existing product details.

2.1.3 Non-functional Testing

Non-functional testing is performed to evaluate the overall quality of the DobiDemo system, including usability, performance, security, and reliability of the system.

— **Usability Testing**
This testing checks whether the system is easy to use and understand. It ensures that users can navigate between modules such as Product, Supplier, Customer, and Cash Flow without difficulty. It also verifies that buttons, forms, and error messages are clear and user-friendly.

— **Performance Testing**
This testing evaluates how fast and responsive the system is during operation. It ensures that pages such as product lists and transaction records load within an acceptable time and that the system can handle multiple operations without slowing down.

— **Security Testing**
This testing ensures that the system protects user data and prevents unauthorized access. It verifies that only valid users can log in and that incorrect login attempts are properly handled.

— **Reliability Testing**

This testing checks whether the system runs consistently without crashing or producing errors during normal usage. It ensures that all operations, such as adding or updating data, work smoothly.

— **Compatibility Testing**
This testing ensures that the system works correctly across different environments. In this project, the system is tested using Google Chrome on a Windows operating system to confirm proper functionality.

2.2 Test Levels

Integration tests are carried out to verify that different modules within the system interact properly with each other This is important because the DobiDemo system consists of multiple interconnected components, such as sales, customers, and cash flow. For example, when a sales transaction is recorded, the system should automatically update the customer's balance and reflect the changes in the cash flow module. Therefore, this level of testing ensures that the data is transferred accurately between modules without any inconsistency.

System testing is performed to evaluate the system as a whole. At this stage, all functionalities are tested together in an end-to-end manner to ensure that the system operates smoothly. For example, processes such as adding a customer, recording a sale, processing payment, and viewing the dashboard are tested as a complete workflow. As a result, this helps confirm that the entire system behaves as expected under normal operating conditions.

Acceptance Testing is conducted to determine whether the system meets user requirements and supports real-world usage. In this project, testing is performed from the perspective of the system owner, who is the main user of the system. Thus, this level of testing ensures that the system is not only technically correct but also practical, usable, and aligned with business needs.

2.3 Test Environment

2.3.1 Setup

The testing environment will be conducted using laptops or desktop computers running Windows operating systems with stable internet connection. NetBeans will be used as the main development platform for managing project files and testing activities. In addition, Apache Tomcat will be used as the application server to deploy and run the system. Furthermore, MySQL will be used to manage the system database and test data. Lastly, GitHub will be used for version control, collaboration, and project submission.

2.3.2 Tools

Tool	Purpose
NetBeans	Developing and managing source code and testing files
Apache Tomcat	Deploying and running the web application
MySQL	Database management and test data storage
Google Chrome	Testing user interface and user interaction

2.4 Test Data

2.4.1 Product Module Test

Test ID.	Description	Test Data	Expected Outcome	Type
P-01	Create new product	Name : Detergent Powder, Unit : kg, Price : 4.00	Product is successfully added and stored in database	Positive
P-02	Add another valid product	Name : Fabric Softener, Unit : bottle, Price : 5.00	Product is successfully added	Positive
P-03	Search product by name	Keyword : Detergent	Matching product(s) displayed	Positive
P-04	View product list	Sort by unit price	Products displayed in ascending/descending order	Positive
N-01	Add product with empty name	Name : "", Unit : kg, Price : 5.00	System shows validation error "Product name is required"	Negative
N-02	Add product with negative price	Name : Soap, Unit : pcs, Price : -3.00	System rejects input and shows error message	Negative
N-03	Search non-existing product	Keyword : 123456	System displays "No product found"	Negative

2.4.2 Supplier Module Test

Test ID	Description	Test Data	Expected Outcome	Type
P-01	Add valid supplier	Name : UMT Supply, Phone : 0123456789, Balance : 150	Supplier successfully added	Positive
P-02	View supplier details	Supplier ID : 1	Supplier details and transaction history displayed	Positive
P-03	Sort suppliers by name	Sort A-Z	Supplier list sorted correctly	Positive
N-01	Add supplier with invalid phone	Phone : ABCD1234	System shows validation error	Negative
N-02	Add supplier with empty name	Name : "", Phone : 0123456789	System rejects input	Negative
N-03	Search non-existing supplier	Keyword : Unknown	System shows "No supplier found"	Negative

2.4.3 Customer Module Test

Test ID	Description	Test Data	Expected Outcome	Type
P-01	Add valid customer	Name : Umairah, Phone : 01136651302, Balance : 0	Customer successfully added	Positive
P-02	View customer details	Customer ID : 2	Customer info and sales history displayed	Positive
P-03	System categorizes customer	Sales count > threshold	Customer category updated automatically	Positive
P-04	Sort by account balance	Sort descending	Customers sorted correctly	Positive
N-01	Add customer with invalid phone	Phone : 123ABC	System rejects input	Negative
N-02	Add customer with empty name	Name : "", Phone : 0131234567	Validation error shown	Negative
N-03	Search invalid customer	Keyword : Unknown	"No customer found" displayed	Negative

3
Test Execution

This part outlines detailed descriptions of the entry and exit criteria, test schedule, and deliverables, to ensure that all aspects of the test execution phase are clearly defined and understood by all stakeholders.

3.1 Entry Criteria

Entry criteria define the prerequisites that must be met before testing can commence.

— **Test Environment :** The test environment is set up, configured, and ready for testing.
— **System Build :** A stable version of the DobiDemo system is available for testing.
— **Test Data :** Required test data is prepared, validated, and available.
— **Test Cases :** All test cases are designed, reviewed, and approved.
— **Resources :** Test Analyst and Test Engineer are assigned and ready to perform testing.

3.2 Exit Criteria

Exit criteria define the conditions that must be met for the testing phase to be considered complete.

— **Test Case Execution :** All planned test cases are executed and results are recorded.
— **Defect Status :** All major defects are fixed and verified. Minor defects are documented.
— **Test Results :**All test results are reviewed and finalized.
— **System Functionality :** The system operates correctly according to the specified requirements.

3.3 Test Schedule

Provide a timeline for testing activities, including milestones and deadlines.

Milestone Task - Phased Plan	Effort (Days)	Start Date	End Date
Test Plan Sign off	1 09/04/26	09/04/26	
Test Case Design	9 10/04/26	19/04/26	
Test Environment Setup	18 20/04/26	07/05/26	
Test Execution	26 08/05/26	05/06/26	
Defect Fixing	5 06/06/2 6	11/06/26	

3.4 Deliverables

This section specifies the various documents and artifacts that will be produced and delivered during the testing process. These deliverables provide valuable documentation and evidence of the testing activities and outcomes.

Test Plan Document

— **Description :** A document describing the testing strategy, scope, and approach.
— **Purpose :** To provide guidance for the testing process.

— **Responsibility :** Project Manager

Test Cases Document

— **Description :** A document containing all test cases for system modules.
— **Purpose :** To ensure all system functions are tested.
— **Responsibility :** Test Analyst, Test Engineer

Test Execution Report

— **Description :** A report showing test execution results and status.
— **Purpose :** To monitor testing progress.
— **Responsibility :** Test Engineer

Defect Report

— **Description :** A document listing all identified defects.
— **Purpose :** To track and manage issues.
— **Responsibility :** Test Analyst